Power Supply

Communications

Wind Sensor

Temperature PT1000 Sensor

File: parameteo_hwrev_h_pwr.kicad_sch File: parameteo_hwrev_h_comms.kicad_sch File: parameteo_hwrev_h_wind.kicad_sch File: parameteo_hwrev_h_pt1000.kicad_sch

Resistors with reference in form of RxyVn where
x and optional y – letters
n – number
are used to configure assembly variants of the pcb.
All resistors with the same x and (optionally) y forms the same variant and all of them have to be placed.
All of them are value of 0 Ohm if not specified otherwise

Please note that some variants are mutually exclusive. You has to make a choice
between one or another. Otherwise the controller will be damaged.

REVn – U7 op amp bypass. In such case
U7 (LMV385), R15, R21, R18, R22 schall not be fitted
RFVn – UART1 on TTL level available on RJ45 jack

(c) Bielsko – Biala 2020, 2021, 2022, 2023, 2024, 2026
tel: +48 660 43 44 46
email: mateusz.lubecki@outlook.com
ML Elektronika Mateusz Lubecki

Sheet: /
File: parameteo_hwrev_h.kicad_sch

Title: ParaMETEO – PV powered, full odu, APRS WX station controller

Size: A3	Date: 2026-07-01	**Rev: H**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 1/5

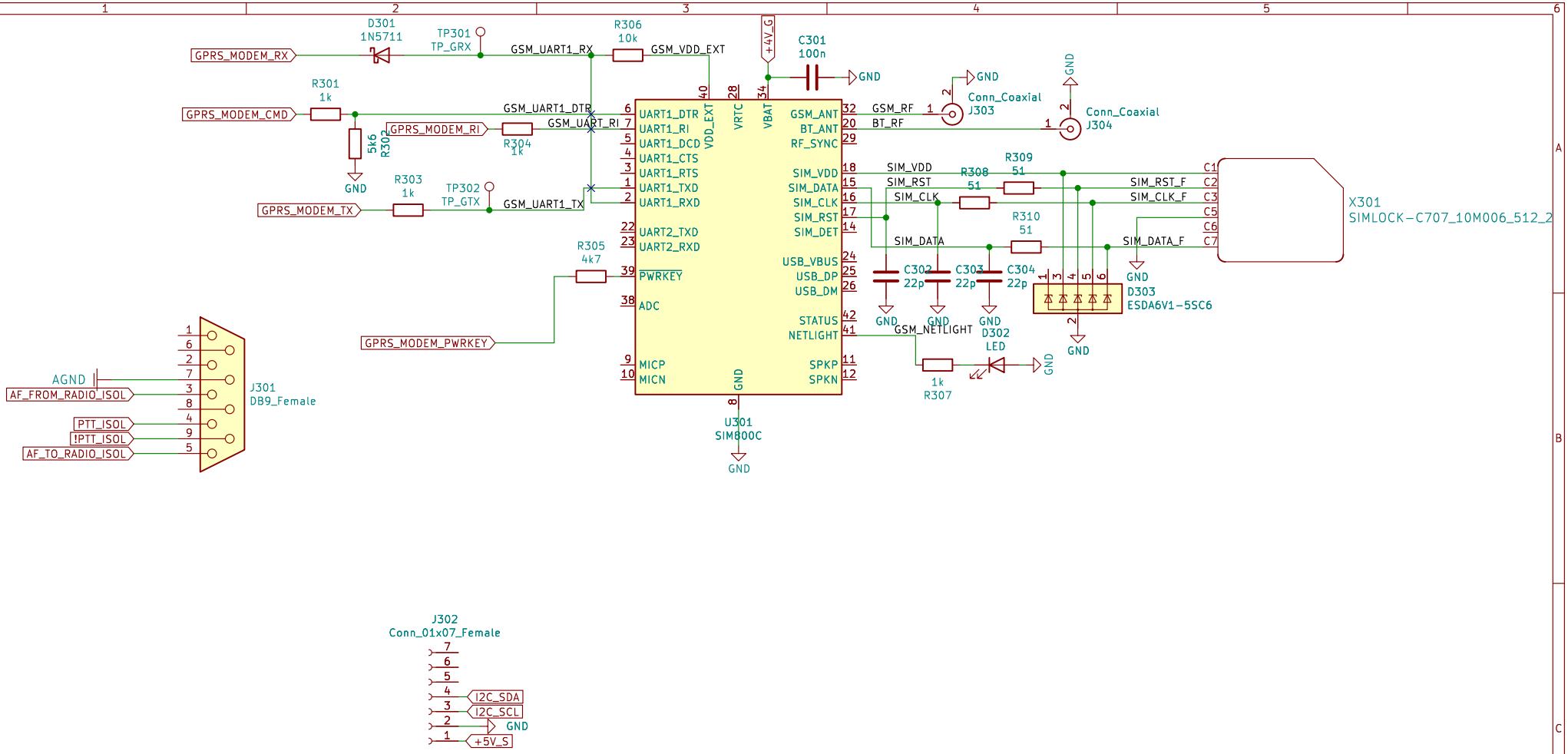
Q204
BSL207SP
FL203
NFE31PT222Z1E9L
VBATT_PROT
VBATT_SW_M
R208 10k
R206 2k2
UC_CNTRL_VM
Q202
BC846
TP202
TP_CNTRL_VBATT_M
10k
R235
GND
Q203
BSL207SP
FL202
NFE31PT222Z1E9L
VBATT_SW_R
C206
47u
R207 10k
R205 10k
CNTRL_VBATT_R
Q201
BC846
TP201
TP_CNTRL_VBATT_R
U204
MAX1837EUT33+T
SHDN
IN
FB
OUT
LX
GND
MAX33_OUT
MAX33_LX
MAX33_IN
MAX33_FB
C212 330n
C213 10u
L206 22u
VISHAY 293D107X96R3C2TE3
100uF, 6.3VDC, SMC
D205
STPS0530Z
C227 47uF
test
68uF
C230 330n
+3V3_C
U205
LP2985-3.3
VIN
VON/OFF
VOUT
BP
LP33_VDDA_IN
LP33_VDDA_OUT
RBV201 DNP
RAV201 0
RBV202 DNP
VDDA_UNFILT
C214 DNP
C215 DNP
C220 DNP
C225 10u
L207 100n
C231 100n
C233 10u
VDDA
RBBV201 DNP
RBAV201 DNP
If MCP1755T-5002E is used C40 shall not be fitted
UC_CNTRL_VS
UC_CNTRL_VC
UC_CNTRL_VG
GSM_PWR_CNTRL
R232 620
R233 620
R234 620
C239 100n
C242 100n
CNTRL_VBATT_S
CNTRL_VBATT_C
CNTRL_VBATT_G
R237 OR
R236 OR
CNTRL_VBATT_R
NLV10MTC4R7 or
NLCV32T-4R7M-EF or
TDK NLCV32T-1ROM-EF
L201 1u
U202
LM2596S-ADJ
VIN
FB
ON/OFF
GND
OUT
LM_VR_VIN
LM_VR_FB
For +5.15V outpu place
1k5 as R220 and
4k7 as R219
L12 FERROCORE DE1207-47
R215 1k
R214 5k1
L204 47u
D201
MBRS340
C202 10u
C204 10u
C215 100n
C216 220u
+7V5_R
R210 10k
R209 10k
Q205
BC846
NLV10MTC4R7 or
NLCV32T-4R7M-EF or
TDK NLCV32T-1ROM-EF
L208 1u
U208
LM2596S-ADJ
L5 FERROCORE DE1209-100
LM_VG_VIN
LM_VG_FB
R236 1k
C226 10u
C232 10u
R227 10k
R220 10k
R231 2k2
L210 100u
D210
MBRS340
C244 100n
C246 470u
+4V_G
TP205
TP_CNTRL_VBATT_G
R222 100k
C234 100n
Q208
BC846
Q207
BSL207SP
RCV202 10k
U203
MAX1837EUT50+T
MAX5_OUT
MAX5_LX
MAX50_IN
L205 22u
C210 100n
C211 4u7
R239 100k
C222 1u
RCV203 0
C229 68u
test
68uf
+5V_S
D204
STPS0530Z
RCV201 10k
Q206
BC846
TP203
TP_CNTRL_VBATT_S
22uH inductors: Viking NLV10KTC220
(SMD; 1210; 22uH; 300mA; 770m; Q15)
+5V_C
+5V_S
+4V_G
SW_M201
LED
5V_S201
LED
5V_S202
LED
R201 2k2
R202 2k2
R204 2k2
Resistors with reference in form of RxyVn where
x and optional y – letters
n – number
are used to configure assembly variants of the pcb.
All resistors with the same x and (optionally) y forms the same variant and all of them have to be placed.
All of them are value of 0 Ohm if not specified otherwise
Please note that some variants are mutually exclusive. You has to make a choice
between one or another. Otherwise the controller will be damaged.
RAV – Micro analog domain (VDDA) is powered from +3V3_C
RBV – Micro analog domain (VDDA) is powered from it's own regulator
RBAV – Supply for micro analog domain can be controled
RBBV – Supply for micro analog domain is permamently powered up
RCV – +5V_S is delivered by MAX18375
RDV – +5V_S is delivered by LP2985
U201
LP2985-5.0
VBATT_PROT
CNTRL_VBATT_S
RDV201 DNP
ON/OFF
BP
LP5S_VOUT
RDV202 DNP
C205 330n
U207
LP2985-5.0
LP5C_VIN
LP5C_VOUT
C221 330n
C224 10u
10k
R221
R240 100k
C236 330n
C237 330n
C240 47u
CNTRL_VBATT_C
TP206
TP_CNTRL_VBATT_C
If MCP1755T-5002E is used C40 shall not be fitted
PV connector for cabling harness
XY2500F-AV(5.0)-2P XINYA
TP213
TP_VPV_FLTRED
TP211
TP_VPV_PFLTRED
VPV_INPUT
VPV_PFLTRED
VPV_FLTRED
L203 7u6
L202 7u6
PV201
PV
C208 4u7
C201 4u7
C203 47u
C209 100u
D203
MBRA340T3
VIN
C207 4u7
C217 4u7
R212 1M4
R213 174k
D202
1N4148W
radiator
FISCHER ELEKTRONIK SK09-20SA220
U206
LT3652EMSE
SHDN
VIN_REG
CHRG
FAULT
TIMER
SW
BOOST
SENSE
BAT
NTC
VFB
!SHDN
VIN_REG
TP_VIN_REG
TP204
R216 470k
R217 100k
LD_CHRG
LED
CHRG201
LED
FAULT201
LD_FLT
C223 4u7
R218 5k1
R219 5k1
R223 10k
TP_VFB
TP207
Rfb3
R224 20k
R225 1M
Rfb1
V_float 14.2V
R226 300k
Rfb2
D208
MBRA340T3
D206
1N4148W
D207
BZX84C6V2L
C235 1u
L209 22u
TP208
TP_SENSE
R228 OR1
R228 VIKING CS12FTFSR100
C218 4u7
C241 4u7
B3100BF-13
D211
This should be MBRS340
but it could be B340B
although it is smaller
VBATT_PROT
TP209
TP_VBATT_PROT
C243 470u
R229 33k
R230 3k3
VBAT_MEAS
TP210
TP_VBAT_MEAS
J201
Screw_Terminal_01x03
VBATT_SW_R
+7V5_R
plug EB5A-03-A ADAM TECH
socket EBRA-03-A ADAM TECH or
15EDGVC-3.5-03P-14-00A(H)
DEGSON ELECTRONICS
EMI shield for power supply 4020.16 TEKO
BATT201
BATTERY
VBATT_IN
FL205
NFE31PT222Z1E9L
VBATT
C228 100uF
VISHAY
MAL213660221E3
TP212
TP_VBAT_PROT1
TP214
TP_VBAT_PROT2
H202
MountingHole
H203
MountingHole
H201
MountingHole_Pad
R203 0
H205
MountingHole_Pad
H206
MountingHole_Pad
H207
MountingHole_Pad
H208
MountingHole_Pad

(c) Bielsko – Biała 2020, 2021, 2022, 2023, 2024, 2026
tel: +48 660 43 44 46
email: mateusz.lubecki@outlook.com
ML Elektronika Mateusz Lubecki

Sheet: /Power Supply/
File: parameteo_hwrev_h_pwr.kicad_sch

Title: ParaMETEO – PV powered, full odu, APRS WX station controller

Size: A3	Date: 2026-07-01	**Rev: H**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 2/5



Resistors with reference in form of RxyVn where
x and optional y – letters
n – number
are used to configure assembly variants of the pcb.
All resistors with the same x and (optionally) y forms the same variant and all of them have to be placed.

Please note that some variants are mutually exclusive. You has to make a choice
between one or another. Otherwise the controller will be damaged.

REVn – U7 op amp bypass. In such case
U7 (LMV385), R15, R21, R18, R22 schall not be fitted
RFVn – UART1 on TTL level available on RJ45 jack

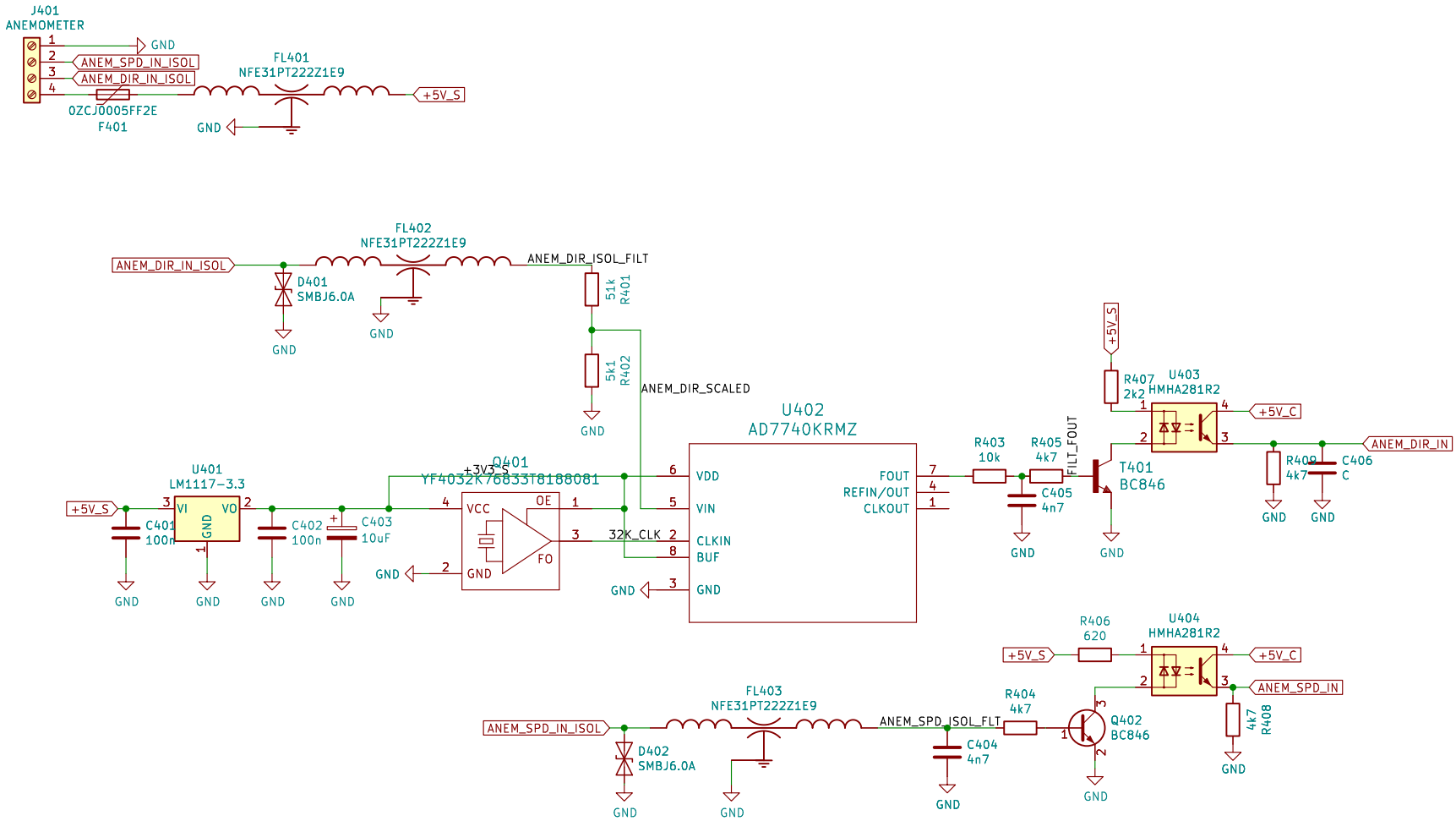
B6B-XH-A JST
https://www.tme.eu/en/details/b6b-xh-a/raster-signal-connectors-2-50mm/jst/

(c) Bielsko – Biala 2020, 2021, 2022, 2023, 2024, 2026
tel: +48 660 43 44 46
email: mateusz.lubecki@outlook.com
ML Elektronika Mateusz Lubecki

Sheet: /Communications/
File: parameteo_hwrev_h_comms.kicad_sch

Title: ParaMETEO – PV powered, full odu, APRS WX station controller

Size: A4	Date: 2026-07-01	**Rev: H**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 3/5

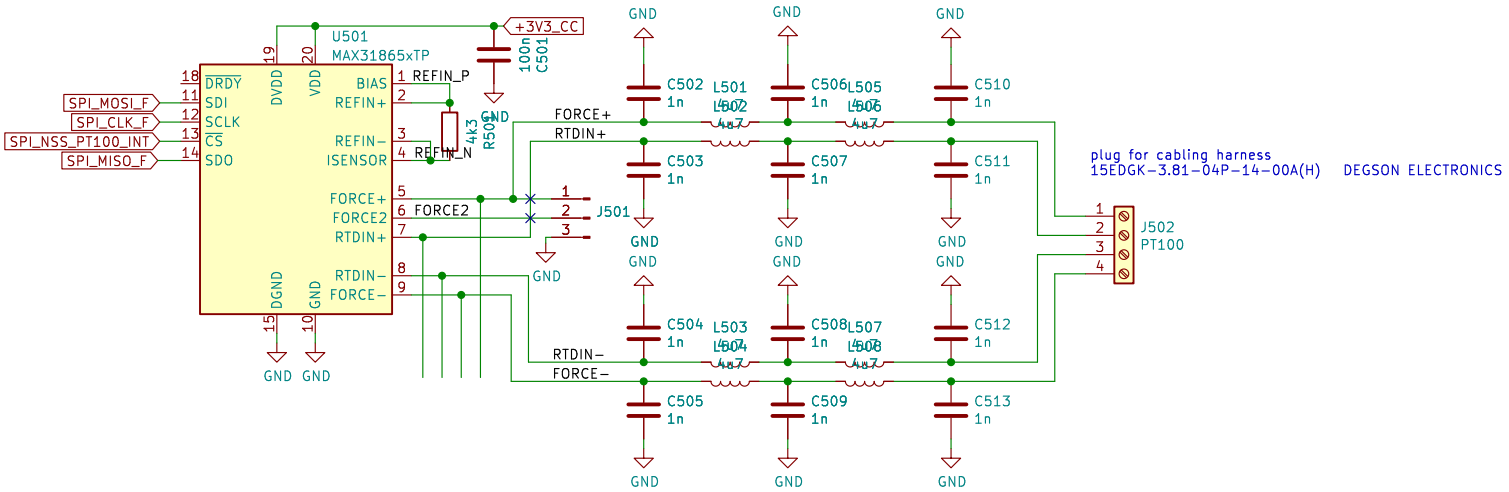


(c) Bielsko – Biala 2020, 2021, 2022, 2023, 2024, 2026
tel: +48 660 43 44 46
email: mateusz.lubecki@outlook.com
ML Elektronika Mateusz Lubecki

Sheet: /Wind Sensor/
File: parameteo_hwrev_h_wind.kicad_sch

Title: ParaMETEO – PV powered, full odu, APRS WX station controller

Size: A4	Date: 2026-07-01	**Rev: H**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 4/5



(c) Bielsko − Biala 2020, 2021, 2022, 2023, 2024, 2026
tel: +48 660 43 44 46
email: mateusz.lubecki@outlook.com
ML Elektronika Mateusz Lubecki

Sheet: /Temperature PT1000 Sensor/
File: parameteo_hwrev_h_pt1000.kicad_sch

Title: ParaMETEO − PV powered, full odu, APRS WX station controller

Size: A4	Date: 2026−07−01	**Rev: H**
KiCad E.D.A. kicad 7.0.11+dfsg−1build4		Id: 5/5